



ST BENEDICT'S COLLEGE

SUBJECT	LIFE SCIENCES PAPER II	DATE	20 NOVEMBER 2023
GRADE	11	MARKS	100
EXAMINER NAME	MRS S. STEGMANN	MODERATOR	MRS B. ZAJAC
CLASS		DURATION	2 HOURS

INSTRUCTIONS:

1. This paper consists of 24 pages (including 2 pages of additional writing space). Please check that your question paper is complete.
 2. Answer all questions on the question paper.
 3. Indicate clearly when the additional writing space has been used to answer a question.
 4. Write in blue or black pen only.
 5. Read all questions carefully and answer in as much detail as possible.
 6. All diagrams should have a clear heading.
 7. It is in your own interest to write legibly and to present your work neatly.
-

QUESTION	POSSIBLE MARK	MARK ACHIEVED
1	30	
2	30	
3	40	
TOTAL: 100		

SECTION A**Question 1:**

Read the information below and answer the questions that follow.

Overview

Typhoid fever is a life-threatening infection caused by the bacterium *Salmonella* Typhi. It is usually spread through contaminated food or water. Once *Salmonella* Typhi bacteria are ingested, they multiply and spread into the bloodstream.

Urbanization and climate change have the potential to increase the global burden of typhoid. In addition, increasing resistance to antibiotic treatment is making it easier for typhoid to spread in communities that lack access to safe drinking water or adequate sanitation.

Symptoms

Salmonella Typhi lives only in humans. Persons with typhoid fever carry the bacteria in their bloodstream and intestinal tract. Symptoms include prolonged high fever, fatigue, headache, nausea, abdominal pain, and constipation or diarrhoea. Some patients may have a rash. Severe cases may lead to serious complications or even death. Typhoid fever can be confirmed through blood testing.

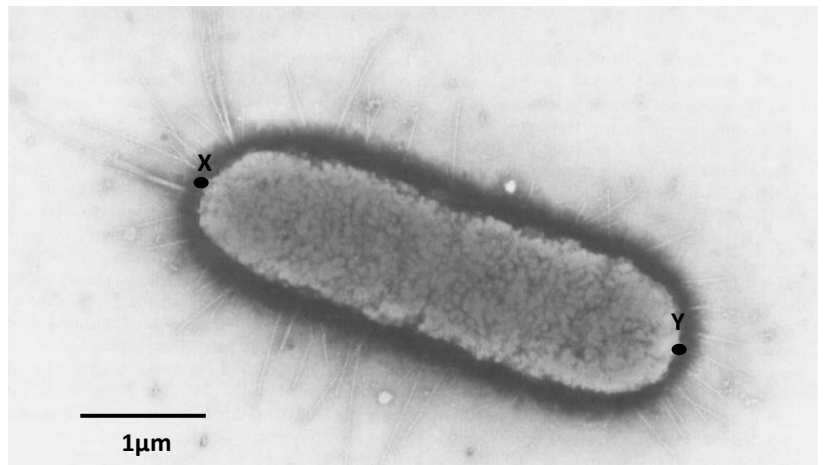


Image 1: *Salmonella typhi* bacterium

Epidemiology, risk factors and disease burden

Improved living conditions and the introduction of antibiotics resulted in a drastic reduction of typhoid fever morbidity and mortality in industrialized countries. However, the disease continues to be a public health problem in many developing areas of the WHO African, Eastern Mediterranean, South-East Asia and Western Pacific Regions.

As of 2019 estimates, there are 9 million cases of typhoid fever annually, resulting in about 110 000 deaths per year. Typhoid risk is higher in populations that lack access to safe water and adequate sanitation, and children are at highest risk.

Treatment

Typhoid fever can be treated with antibiotics. Antimicrobial resistance is common with the likelihood of more complicated and expensive treatment options required in the most affected regions. Even when the symptoms go away, people may still be carrying typhoid bacteria, meaning they can spread it to others, through shedding of bacteria in their faeces.

It is important for people being treated for typhoid fever to do the following:

- Take prescribed antibiotics for as long as the doctor has prescribed.
- Wash their hands with soap and water after using the bathroom and avoid preparing or serving food for other people. This will lower the chance of passing the infection on to someone else.
- Have their doctor test to ensure that no *Salmonella* Typhi bacteria remain in their body.

1.1. How is Typhoid fever usually spread? (2)

1.2. Name TWO common symptoms of Typhoid fever. (2)

1.3. Why would a person most likely suffer from dehydration after getting Typhoid fever? (1)

1.4. By what asexual process do bacteria reproduce? (1)

1.5. Calculate the annual percentage of people that die after getting Typhoid fever. (2)

1.6. Explain why Typhoid fever is more common in rural, underdeveloped areas. (2)

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- 1.7. Why is Typhoid fever typically treated with antibiotics? (1)
-
- 1.8. How do you think a person is tested for Typhoid fever? (1)
-
- 1.9. Classify *Salmonella typhi* according to its shape (Image 1). (1)
-
- 1.10. Write down the LETTER /S of the statements you imagine to be TRUE regarding the structure of *Salmonella typhi* bacteria:
- A. Bacteria contain waxy cuticles.
 - B. Bacteria lack flagella.
 - C. Bacteria contain many pili.
 - D. Bacteria release harmful toxins.
- _____ (2)
- 1.11. Do we classify bacteria as prokaryotes or eukaryotes? (1)
-
- 1.12. Provide a reason for your answer to QUESTION 1.11. (1)
-
-
- 1.13. Using the information in Image 1, calculate the true size of a *Salmonella typhi* bacterium. (4)

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1.14. *E. coli* is a bacteria found in the human gut.

Name and explain the relationship found between *E.coli* and humans.

(3)

1.15. Tabulate TWO differences between viruses and bacteria.

(6)

Total Question 1: 30 marks

Question 2

Read through the information on Malaria below and answer the questions that follow.

Malaria is a life-threatening disease primarily found in tropical countries. It is both preventable and curable. However, without prompt diagnosis and effective treatment, a case of uncomplicated malaria can progress to a severe form of the disease, which is often fatal without treatment.

Malaria is not contagious and cannot spread from one person to another; the disease is transmitted through the bites of female *Anopheles* mosquitoes. Five species of parasites can cause malaria in humans and 2 of these species – *Plasmodium falciparum* and *Plasmodium vivax* – pose the greatest threat. There are over 400 different species of *Anopheles* mosquitoes and around 40, known as vector species, can transmit the disease.

This risk of infection is higher in some areas than others depending on multiple factors, including the type of local mosquitoes. It may also vary according to the season, the risk being highest during the rainy season in tropical countries.

Nearly half of the world's population is at risk of malaria. In 2021, an estimated 247 million people contracted malaria in 85 countries. That same year, the disease claimed approximately 619 000 lives.

Some people are more susceptible to developing severe malaria than others. Infants and children under 5 years of age, pregnant women and patients with HIV/AIDS are at particular risk.

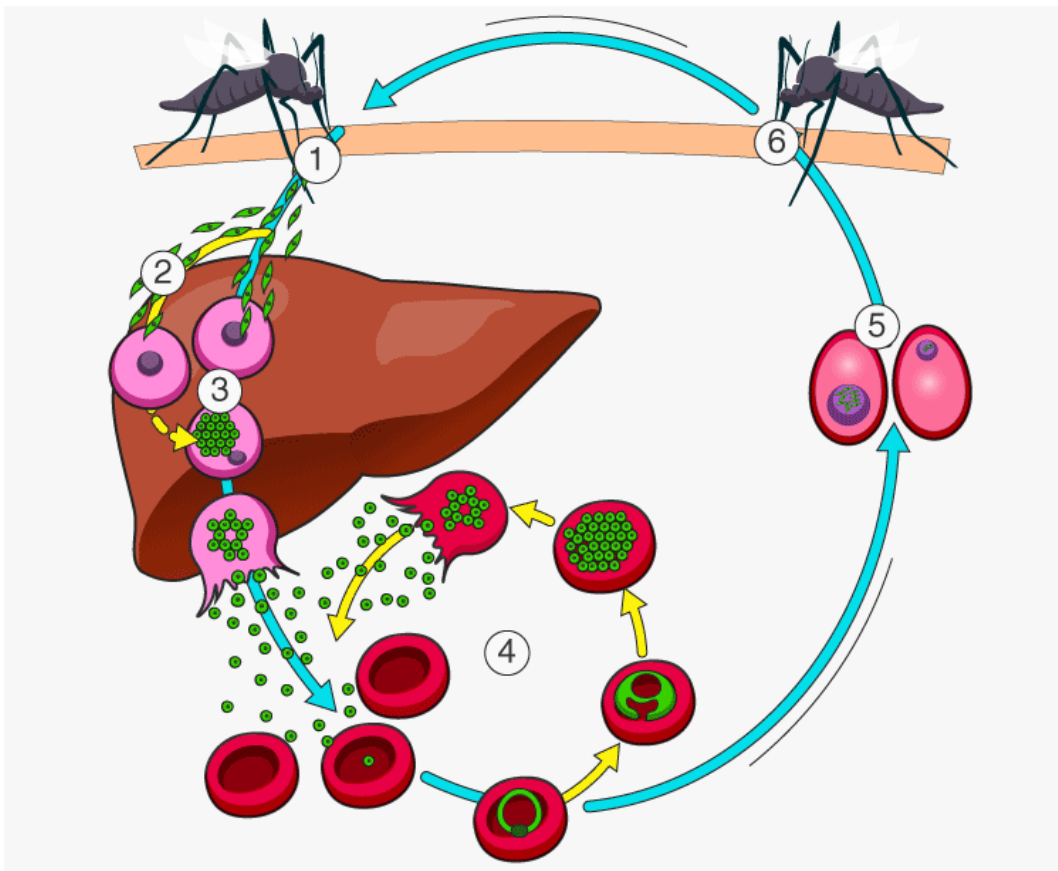


Image 2: Life cycle of malaria

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2.1. To what group of microorganisms does *plasmodium* belong? (1)

2.2. How many different species of parasite can cause malaria in humans? (1)

2.3. Which are the two most dangerous species of malaria parasites to humans? (2)

2.4. Name TWO climatic conditions that increase the risk of infection. (2)

2.5. Why is the female *Anopheles* mosquito known as a vector? (2)

2.6. Calculate the percentage of people died of malaria in 2021? (2)

2.7. Why do you think young children and people suffering with HIV/AIDS are more likely to develop severe malaria? (4)

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2.8. Using the numbers 1-6 on Image 2, describe the life cycle of malaria.

(6)

Number	Description
1	
2	
3	
4	
5	
6	

2.9. The table below shows the number of recorded cases and deaths of malaria in South Africa from 2015-2019.

Year	Total recorded cases	Total number of deaths
2015	12 350	8
2016	5 020	40
2017	30 000	22
2018	18 900	120
2019	14 900	58

2.9.1. Which year recorded the highest number of cases?

(1)

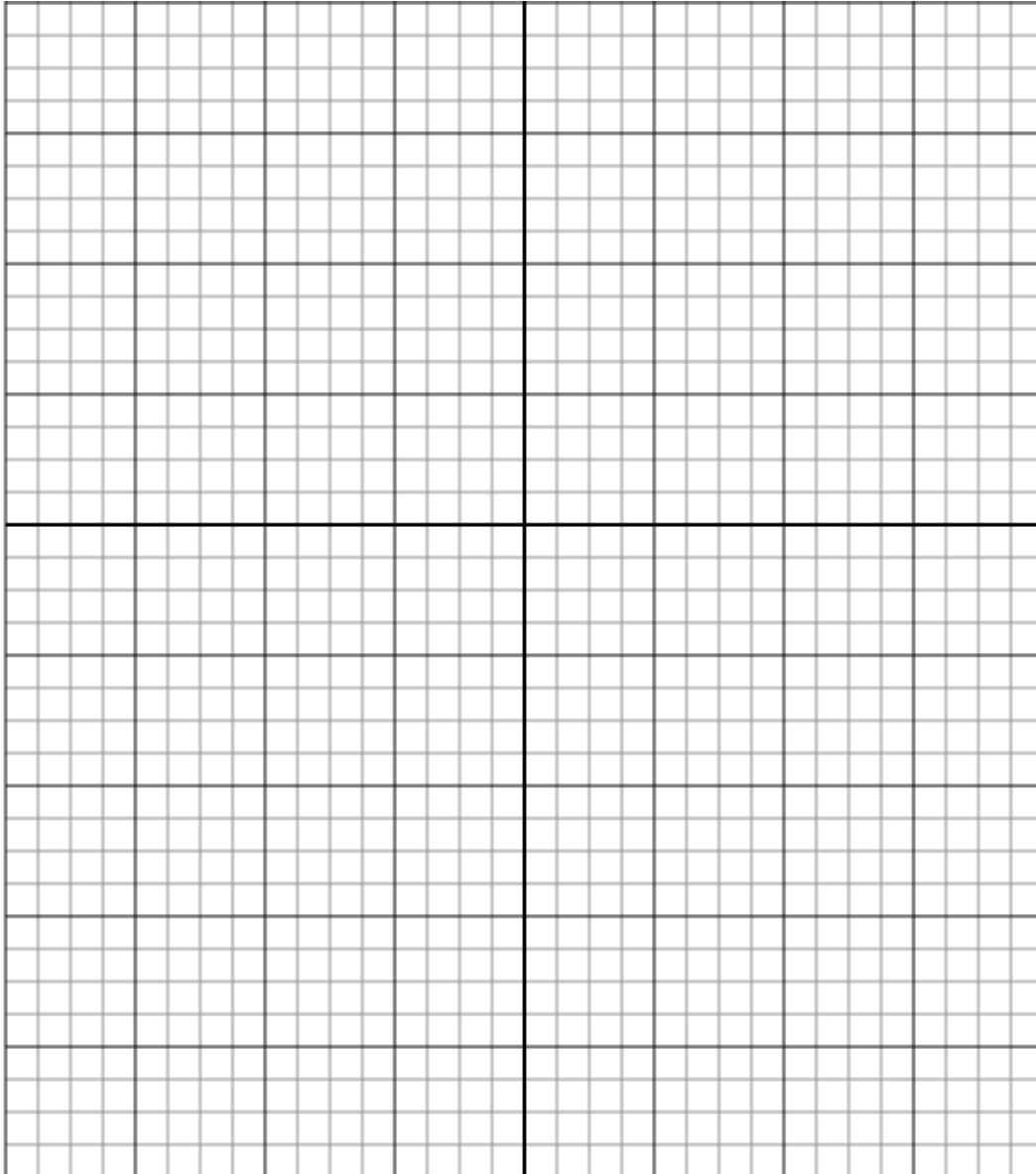
2.9.2. Which year recorded the highest percentage of deaths?

(2)

2.9.3. Draw a bar graph showing the total number of deaths from 2015-2019 below.

(7)

Heading: _____



Total Question 2: 30 marks

60 marks

SECTION B**Question 3**

Covid-19 vaccines should be made compulsory.

Using the source material provided as well as your own knowledge, discuss your opinion on the above statement in the form of a 2 ½ - 3 page essay.

40 marks

To answer this question, you are expected to:

- Read the source material carefully and present a debated argument to present your point of view.
- Select relevant information from Sources A to G below.
- It is important to integrate your own relevant biological knowledge.
- Take a definite stand on the question and arrange the information to best develop your argument.
- Write in a way that is scientifically appropriate and communicates your point of view clearly.
- Provide a clear plan of your essay before you start writing. Note that the plan will be marked as part of the assessment of this question.
- You are to provide **8 points** of evidence to back up your opinion of the above statement.

SOURCE A: How do vaccines work?

Germes are all around us, both in our environment and in our bodies. When a person is susceptible and they encounter a harmful organism, it can lead to disease and death.

The body has many ways of defending itself against **pathogens** (disease-causing organisms). Skin, mucus, and cilia (microscopic hairs that move debris away from the lungs) all work as physical barriers to prevent pathogens from entering the body in the first place. When a pathogen does infect the body, our body's defenses, called the immune system, are triggered and the pathogen is attacked and destroyed or overcome.

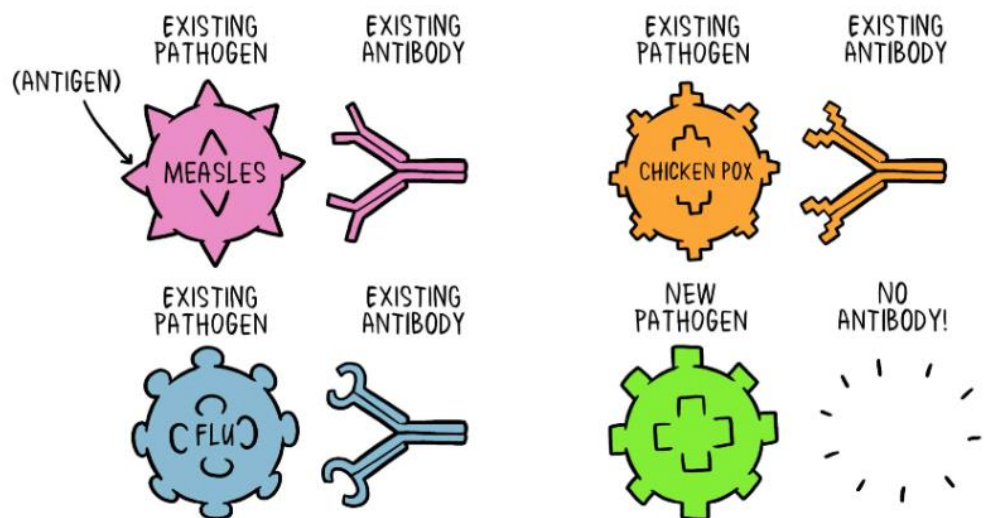
The body's natural response

A pathogen is a bacterium, virus, parasite or fungus that can cause disease within the body. Each pathogen is made up of several subparts, usually unique to that specific pathogen and the disease it causes. The subpart of a pathogen that causes the formation of antibodies is called an antigen. The antibodies produced in response to the pathogen's antigen are an important part of the immune system. You can consider antibodies as the soldiers in your body's defense system. Each antibody, or soldier, in our system is trained to recognize one specific antigen. We have thousands of different antibodies in our bodies. When the human body is exposed to an antigen for the first time, it takes time for the immune system to respond and produce antibodies specific to that antigen.

In the meantime, the person is susceptible to becoming ill.

Once the antigen-specific antibodies are produced, they work with the rest of the immune system to destroy the pathogen and stop the disease. Antibodies to one pathogen generally don't protect against another pathogen except when two pathogens are very similar to each other, like cousins.

Once the body produces antibodies in its primary response to an antigen, it also creates antibody-producing memory cells, which remain alive even after the pathogen is defeated by the antibodies. If the body is exposed to the same pathogen more than once, the antibody response is much faster and more effective than the first time around because the memory cells are at the ready to pump out antibodies against that antigen.

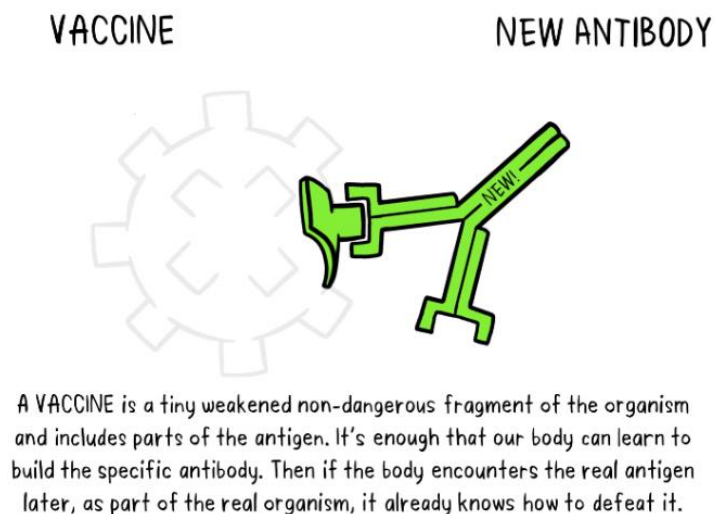


When a new pathogen or disease enters our body, it introduces a new antigen. For every new antigen, our body needs to build a specific antibody that can grab onto the antigen and defeat the pathogen.

This means that if the person is exposed to the dangerous pathogen in the future, their immune system will be able to respond immediately, protecting against disease.

How vaccines help

Vaccines contain weakened or inactive parts of a particular organism (antigen) that triggers an immune response within the body. Newer vaccines contain the blueprint for producing antigens rather than the antigen itself. Regardless of whether the vaccine is made up of the antigen itself or the blueprint so that the body will produce the antigen, this weakened version will not cause the disease in the person receiving the vaccine, but it will prompt their immune system to respond much as it would have on its first reaction to the actual pathogen.



Some vaccines require multiple doses, given weeks or months apart. This is sometimes needed to allow for the production of long-lived antibodies and development of memory cells. In this way, the body is trained to fight the specific disease-causing organism, building up memory of the pathogen so as to rapidly fight it if and when exposed in the future.

Herd immunity

When someone is vaccinated, they are very likely to be protected against the targeted disease. But not everyone can be vaccinated. People with underlying health conditions that weaken their immune systems (such as cancer or HIV) or who have severe allergies to some vaccine components may not be able to get vaccinated with certain vaccines. These people can still be protected if they live in and amongst others who are vaccinated. When a lot of people in a community are vaccinated the pathogen has a hard time circulating because most of the people it encounters are immune. So the more that others are vaccinated, the less likely people who are unable to be protected by vaccines are at risk of even being exposed to the harmful pathogens. This is called herd immunity.

This is especially important for those people who not only can't be vaccinated but may be more susceptible to the diseases we vaccinate against. No single vaccine provides 100% protection, and herd immunity does not provide full protection to those who cannot safely be vaccinated. But with herd immunity, these people will have substantial protection, thanks to those around them being vaccinated.

Vaccinating not only protects yourself, but also protects those in the community who are unable to be vaccinated. If you are able to, get vaccinated.

Throughout history, humans have successfully developed vaccines for a number of life-threatening diseases, including meningitis, tetanus, measles and wild poliovirus. In the early 1900s, polio was a worldwide disease, paralysing hundreds of thousands of people every year. By 1950, two effective vaccines against the disease had been developed. But vaccination in some parts of the world was still not common enough to stop the spread of polio, particularly in Africa. In the 1980s, a united worldwide effort to eradicate polio from the planet began. Over many years and several decades, polio vaccination, using routine immunization visits and mass vaccination campaigns, has taken place in all continents. Millions of people, mostly children, have been vaccinated and in August 2020, the African continent was certified wild poliovirus free, joining all other parts of the world except Pakistan and Afghanistan, where polio has not yet been eradicated.



A vaccine protects an individual...



Community vaccination protects the whole community, even those who can't vaccinate.

Adapted from: <https://www.who.int/news-room/feature-stories/detail/how-do-vaccines-work>

SOURCE B: Safety of vaccines**Safety of COVID-19 vaccines for different groups**

COVID-19 vaccines have been tested in large, randomized controlled trials that include people of a broad age range, all sexes, different ethnicities, and those with known medical conditions. The vaccines have shown a high level of efficacy across all populations. Vaccines have been found to be safe and effective in people with various underlying medical conditions that are associated with increased risk of severe disease. These include high blood pressure; diabetes; asthma; pulmonary, liver or kidney disease; and chronic infections that are stable and controlled.

Those who should consult with a doctor before vaccination include people with a compromised immune system, older people with severe frailty, people with a history of severe allergic reaction to vaccines, people living with HIV, and those who are pregnant or breastfeeding. For more information about vaccine safety for different groups, please see our Q&A on COVID-19 vaccine safety.

Adapted from: <https://www.who.int/news-room/feature-stories/detail/safety-of-covid-19-vaccines>

SOURCE C: Impact of vaccines

The primary benefit of vaccination is that it prevents disease. Immunization is considered one of the greatest public health achievements of the 20th century, and experts agree that immunization is key to staying healthy. In one year, vaccines prevent more than 8 500 child hospitalizations in Colorado, 33 000 deaths in the U.S., and between 2 and 3 million deaths worldwide.

Vaccination has decreased the rate of disease dramatically.

Disease	Baseline 20 th century annual cases	2006 cases	Percentage decrease (%)
Measles	503 282	55	99.9
Diphtheria	175 885	0	100
Mumps	152 209	6 584	95.7
Pertussis (whooping cough)	147 271	15 632	89.4
Smallpox	48 164	0	100
Rubella	47 745	11	99.9
Haemophilus influenza type B, invasive (HiB)	20 000	29	99.9
Polio	16 316	0	100
tetanus	1 314	41	96.9

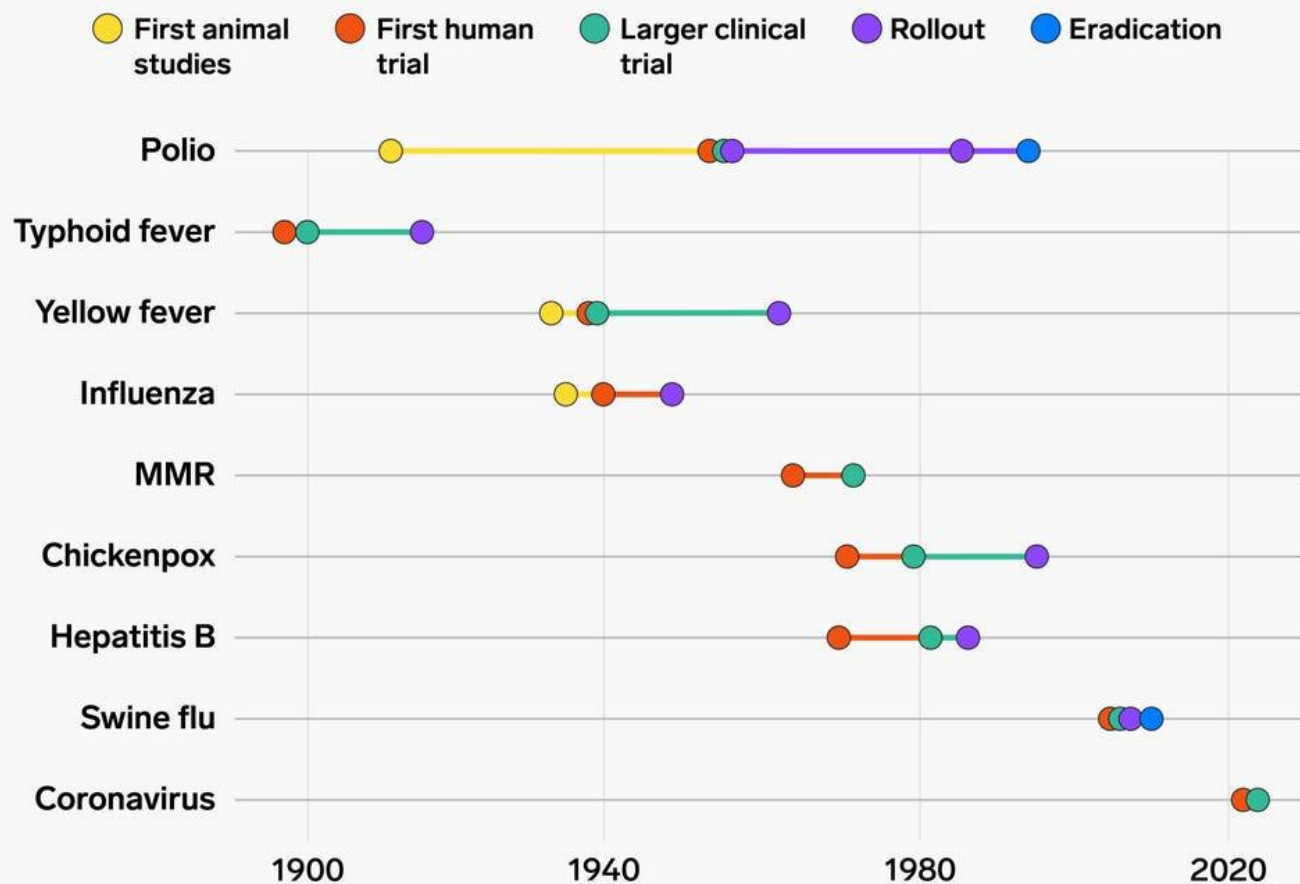
Adapted from: <http://immunizeforgood.com/fact-or-fiction/benefits-vs.-risks>

SOURCE D: Vaccine development

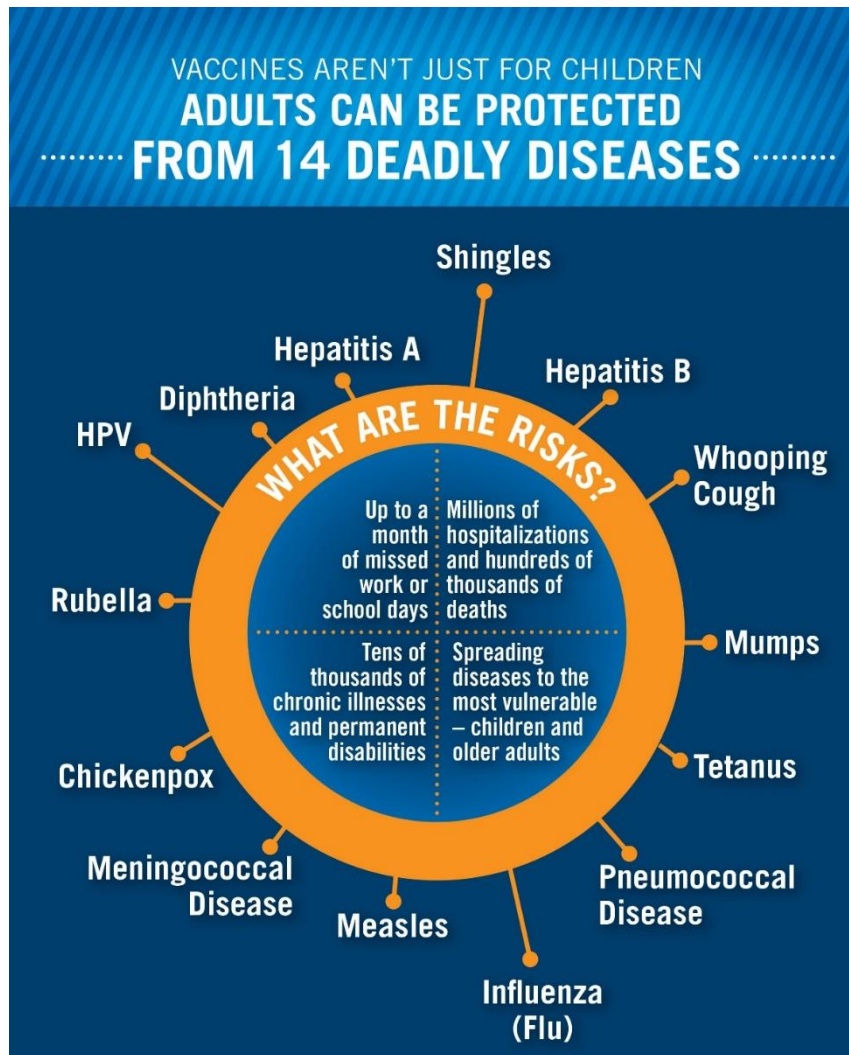
Vaccines undergo a rigorous testing process that can take years, including large Phase III studies that measure safety among thousands of participants. These trials usually follow participants for six months to a year. This is enough of a time frame to assess the safety of a vaccine because any allergic reaction or adverse events happen shortly after vaccination (hours or rarely days).

As with all medicines, vaccines continue to be monitored over time. Most often, long-term data confirms the findings of the clinical trial safety data. In rare instances, an extremely uncommon adverse event can be identified from long-term safety data. If that does happen, the data is evaluated as part of the ongoing monitoring process. Learn more about the research process.

Comparing vaccine development throughout history



Adapted from: <https://www.voicesforvaccines.org/science/vaccinesafety/>

SOURCE E: What vaccines do I need?

Acquired from: <http://www.adultvaccination.org/what-vaccines>

SOURCE F: Mandatory vaccines

Covid vaccinations are already a requirement for public life in many parts of the world. If you are a French doctor, a New Zealand teacher or a Canadian government employee, getting your shots is essential to go to work. Indonesia can deny benefits to people who refuse jabs. Greece is making them compulsory for the over-60s. Austria is set to go further still, with a plan to introduce mandatory vaccinations for all by February. This would not mean Austrians being forcibly injected. There will be medical and religious exemptions. But the bulk of the remaining unvaccinated population face fines for not getting their shots. Here in London, in July anti-lockdown demonstrators took to the streets to demonstrate against a lockdown that had been lifted just hours earlier. The point is, whatever a government does, it will face opposition. Covid restrictions in particular have drawn protests around the world and mandatory vaccinations are a step beyond, say, a mask mandate.

"When it comes to vaccines, people do think very differently," says Vageesh Jain, a public health doctor at the Institute for Global Health at University College London.
"Anything that's administered to them in their body, it's not going to be thought of in the same way, even though academics and others may think theoretically it's just a restriction, people do have this kind of emotive response."

Adapted from: <https://www.bbc.com/news/world-59506339>

SOURCE G: People's opinion

"Had a severe reaction to the flu vaccine so I am very concerned that it could happen with the COVID vaccine"

– 55 year old, female health care worker, Pennsylvania

"Potential side effects and no research on long term effects"

– 23 year old, female, health care worker, Ohio, received vaccine

"I don't believe the FDA is telling the truth. The vaccine is not ready yet and people I know who have taken it are having serious side effects and doctors are covering it up."

– 42 year old, female, North Carolina

KFF COVID-19 Vaccine Monitor: In Their Own Words

"I am scared that the vaccine is not going to do my body well. I think that it will give me some problems."

– 25-year-old woman from Ohio

"I believe that it will ultimately provide the best defense against the disease."

– 40-year-old male health care worker from Florida

Adapted from: https://www.google.com/search?q=people%27s+positive+opinions+on+vaccines&sca_esv

PLANNING:

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, typical of notebook paper. There are no margins, text, or other markings on the page.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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